PRODUCT USER MANUAL

For
Black Sea Rim Current Intensity Index

OMI_CIRCULATION_BOUNDARY_BLKSEA_rim_current_index

Issue 2.0

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Issue	Date	§	Description of Change	Author	Checked by	Validated by
1.0	2023/06/15	all	Creation of the document	E. Jansen		
2.0	2025/05/28	all	Update the time series based on the results from Black Sea Reanalysis version BSe4r1 and s etup of o perational e xtensions	E. Jansen, L. Lima, R. G. de Menezes	R. Lecci	Copernicus Marine Product Management

GLOSSARY AND ABBREVIATIONS

MOI	Mercator Océan International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CIRCULATION_BOUNDARY_BLKSEA_rim_current_index . It provides an estimate of the intensity of the Black Sea R im C urrent starting in 1993 and extended operationally on a yearly basis up to one year before the current year .

The product contains the Black Sea R im C urrent intensity index (BSRCI) calculated from the BLKSEA_MULTIYEAR_PHY_007_004 reanalysis product. The RCI is a dimensionless quantity, expressing the intensity of the R im C urrent in a particular year in relation to climatological average.

The product is available for the Black Sea area and calculated as a time series with yearly frequency. Three time series are provided, corresponding to the BSRCI calculated for the northern R im current, the Southern R im C urrent and the combination of the two areas. More details about the calculation can be found in the corresponding Quality Information Document (QUID) document.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is described in the QUID CMEMS-BLK-OMI-QUID-CIRCULATION-BOUNDARY_rim_current_index V2.0 .pdf (see V).

Information on operational issues on products and services can be found on our **User Notification Service**. If you have any questions, please **contact us**.

I.2. History of changes

2023/06	1st publication of the product: OMI_CIRCULATION_BOUNDARY_BLKSEA_rim_current_index
2025/11	Update the time series based on the results from Black Sea Reanalysis version BSe4r1 and s etup of operational extensions and re-delivery of time series

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	OMI_CIRCULATION_BOUNDARY_BLKSEA_rim_current_index
Geographical coverage	Black Sea
Variables	rim_current_intensity_index for three different areas in the Black Sea: - The northern rim current - The southern rim current - The combination of north and south
Update frequency	yearly
Available time series	From 1993 until one year before the current one [-1 Y]
Delivery mechanism	MDS
Temporal/Spatial resolution	Yearly average
Integration depth (for example..)	surface
Format	NetCDF CF1.7

II.2. Details of the dataset

omi_circulation_boundary_blksea _rim_current_index contains the 1D yearly rim current intensity index calculated from BLKSEA_MULTIYEAR_PHY_007_004
BSRCI [1] Dimensionless index representing the intensity of the Black Sea rim current as a ratio to the interannual mean value. Calculated using the northern and southern rim current. rim_current_intensity_index **BSRCIn [1]** As BSRCI, but calculated using only the northern rim current. rim_current_intensity_index **BSRCIs [1]** As BSRCI, but calculated using only the southern rim current. rim_current_intensity_index

Note: the standard name and units of the rim current intensity index (rim_current_intensity_index) are not yet included in CF conventions since they were created for this first contribution.

II.3. Product System Description

The estimates are derived from the daily mean surface currents of the Black Sea regional reanalysis BLKSEA_MULTIYEAR_PHY_007_004. The rim current intensity uses the mean surface current velocity in a northern and southern area of the Black Sea rim current. The index yearly ratio of the average velocity with the interannual mean. More details can be found in the Quality Information Document (QUID) CMEMS-BLK-OMI-QUID -CIRCULATION-BOUNDARY_rim_current_index V2.0 .pdf (see V).

III FILES NOMENCLATURE AND FORMAT

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

III.1. Nomenclature of files

OMI_CIRCULATION_BOUNDARY_BLKSEA_rim_current_index file nomenclature when downloaded through the Copernicus Marine Service Download Services is based on the following nomenclature:

<datasetname> .nc

Where :

- **<datasetname>** is the dataset reference**.**
-
- **.nc** is the standard NetCDF filename extension.

For example: **omi_circulation_boundary_blksea_rim_current_index .nc**

III.2. File Format

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IVREFERENCES

See the Quality Information Document (QUID):
https://documentation.marine.copernicus.eu/QUID/CMEMS-OMI-QUID-CIRCULATION-bsrci.pdf
https://documentation.marine.copernicus.eu/QUID/CMEMS-BLK-QUID-007-004.pdf